

ZUCHEN LI

📞 217-318-7023 ✉️ zuchenli@umich.edu 🔗 [LinkedIn](#) 🐙 [GitHub](#)

Research Interests: Human-AI Interaction · Context-Aware Intelligent Systems · Ubiquitous Sensing · AI4Science

Education

University of Michigan, Ann Arbor

Aug 2024 – Present

Triple Major in Computer Science, Data Science, & Statistics; Minor in History

3.8/4.0

University of Illinois Urbana-Champaign

Aug 2023 – May 2024

Major in Statistics & Computer Science; Minor in Meteorology

4.0/4.0

Research Experience

Human Centered Computing Lab, University of Michigan

Sept 2025 – Present

Undergraduate Research Assistant

Ann Arbor, Michigan

- **Generative Tutorial** (*Co-first Author; with Profs. Anhong Guo and Xu Wang*): Co-developed a conceptual framework and AR system for live contextualized visual instruction that grounds physical-task goals in users' workspaces and proactively generates workspace-specific goal images and demonstration videos by propagating observed and predicted visual states across task dependencies; co-led a formative evaluation of 176 generated artifacts across 15 tasks and a 24-participant comparative study, showing improved task quality, perceived workspace correspondence, and shorter step-confirmation intervals over pre-authored guidance.
- **MoiréDeform** (*Co-first Author, ACM MobiCom '26 Demo; with Profs. Alanson Sample and Ke Sun*): Developed a low-cost camera-based sensing approach that exploits naturally occurring woven textures as virtual Moiré layers to amplify subtle surface deformation, enabling continuous body-lean interaction and respiratory waveform/rate sensing on an unmodified mesh chair without instrumenting the user or surface; reduced respiratory-rate MAE from 2.07 to 0.71 breaths/min relative to physical-marker tracking.
- **RespiraFrame** (*with Prof. Junyi Zhu*): Contributed to the experimental design and evaluation of an eyeglass-based multimodal respiratory sensing system combining bone-conduction acoustics and electrical impedance tomography (EIT); designed evaluation protocols spanning respiratory behaviors, motion and environmental interference, a three-day semi-wild study, and repeated wear over a four-week longitudinal study.
- **VeilLens** (*Co-first Author, with Prof. Ke Sun*): Developed an agentic visual privacy framework that autonomously discovers, localizes, and selectively edits weak compositional cues exploited by VLMs, supporting granular user control and iterative verification of residual privacy leakage across identity, geolocation, affiliation, and social-relationship risks.

Generative AI for Music and Audio (Advisor: Prof. Herman Dong)

Sept 2025 – Present

Research Assistant

Ann Arbor, Michigan

- Investigating real-time human-GenAI co-creation for controllable music generation, focusing on interaction mechanisms that preserve user agency, expressive control, and iterative creative feedback.
- Prototyping gesture-driven interaction techniques that connect embodied user input to real-time musical control and generative model behavior.

Energy Distance Dimension Reduction (Advisor: Prof. Kerby Shedden)

Jan 2025 – May 2025

Selected Researcher, Undergraduate Research Program in Statistics (URPS)

Ann Arbor, Michigan

- Developed engression and sufficient dimension reduction methods for multilevel, dyadic, and longitudinal wearable data from bone marrow transplant patients, addressing non-i.i.d. structure not handled by existing methodology.
- Implemented sufficient dimension reduction algorithms in JAX with GPU acceleration, achieving approximately 7,000× speedup over the original method for U-statistic computation on large sensor data series.

Lab of Atmospheric and Earth System Sciences, Nanjing University

May 2024 – Aug 2024

Summer Research Intern, Advisor: Prof. Minghuai Wang

Nanjing, China

- Studied statistical machine learning methodologies for integrating real-world sensor observations with simulated atmospheric data, enabling inference and prediction of aerosol-cloud interactions across larger spatial and temporal scales.

E2-E Lab, University of Illinois Urbana-Champaign

Aug 2023 – May 2024

Undergraduate Research Assistant

Champaign, Illinois

- Conducted airflow experiments and processed PIV/camera data using Python, OpenCV, and OpenPIV for flow-field reconstruction and particle tracking; performed CFD simulations in ANSYS Fluent to analyze particle-airflow interactions in complex 3D geometries.

Selected Projects

MagicBaton: Gesture-Based Human–AI Music Interaction Winter 2026

- Built a dual-IMU smart conducting baton that streams 12-axis motion at ~ 50 Hz and integrates real-time edge compute gesture recognition model, jerk-based beat/BPM estimation, MIDI playback, and in-browser Magenta MusicRNN generation.
- Collected a public dataset of approximately 934 recordings across eight gesture classes.

Multimodal Finite State Machine for Adaptive Skill Guidance Winter 2026

- Built a real-time hand-washing guidance system that fuses VLM-based visual cues and YAMNet audio classification with a 12-state finite-state machine, adapting voice-instruction level of detail when users stall or deviate from the procedure.
- Implemented switchable local Qwen3-VL and Gemini backends, sustained-condition transitions, non-blocking TTS, split-screen state visualization, and post-session scoring/logging.

How Fragile Are GUI Agents? A Controlled Perturbation Benchmark for Interface Evolution Winter 2026

- Developed a benchmark grounded in five recurring interface-evolution categories—layout, semantic, visual, structural, and workflow shifts—and generated controlled synthetic websites for systematic analysis of GUI-agent robustness.
- Led project organization and agent experiments using the VisualWebArena harness.

Bayesian MCMC Data Augmentation for Ebola Outbreak Data 2025

- Developed a Bayesian data-augmentation MCMC analysis for partially observed Ebola onset/removal data with censoring.

Additional Experience

Waseda University **June 2025 – Aug 2025**

Summer Intern

Tokyo, Japan

- Researched postwar Japanese literature and history, analyzing literary works in relation to historical memory, identity, and social transformation in postwar Japan.

The Michigan Daily **Aug 2025 – Jul 2026**

Student Reporter, Photographer & Videographer

Ann Arbor, Michigan

- Produced campus and local news coverage through interviews, field reporting, photography, and video, translating on-the-ground observations into written and multimedia stories.

Kendo Club at the University of Michigan

Member

Ann Arbor, Michigan

- Participate in regular kendo practices focused on footwork, striking technique, partner drills, and traditional Japanese swordsmanship.

Michigan Backpacking Club

Member

Ann Arbor, Michigan

- Participate in group hiking and backpacking trips, building outdoor navigation, trip-planning, and teamwork experience.

Illinois Student Organization of Meteorology (ISTORM)

Student Forecaster

Champaign, Illinois

- Student chapter of the American Meteorological Society; focused on severe and hazardous weather forecasting

Honors & Awards

UROP Summer Engineering Fellowship, University of Michigan 2026

University Honors, University of Michigan 2024–2026

Outstanding Research Paper Award, IMMC Greater China Round (top 1.875% of 800+ teams) 2022

Finalist Research Paper Award, IMMC International Round (top 8.75% internationally) 2022

Selected Coursework

EECS 545 Machine Learning · EECS 281 Data Structures and Algorithms · STATS 470 Design of Experiment · EECS 442 Computer Vision · EECS 484 Database Management System · STATS 571 Causal Inference · EECS 498 Machine Learning for Computational Sensor in Mobile/Wearable IoT System · PAT 564 Generative AI for Music and Audio · Math 451 Real Analysis